



Data Article

IC4SD-Wood-Eucalyptus: A macroscopic transverse-section image dataset with metadata, split manifests, and leakage-audit reports for *Eucalyptus* wood identification



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ARTICLE INFO

Article history:

Received 18 May 2026

Revised 8 July 2026

Accepted 30 July 2026

Available online 3 August 2026

Keywords:

Eucalyptus

Wood identification

Macroscopic imaging

Image dataset

Data leakage

Perceptual hashing

Split manifests

Reproducible machine learning

ABSTRACT

IC4SD-Wood-Eucalyptus is a macroscopic transverse-section image dataset for fine-grained *Eucalyptus* wood identification. The dataset contains 2910 images from 86 physical wood specimens collected across two acquisition periods (2023–2024), acquired at 50× magnification from eight classes: seven *Eucalyptus* species (*E. camaldulensis*, *E. cladocalyx*, *E. deglupta*, *E. diversicolor*, *E. grandis*, *E. microcorys*, and *E. saligna*) and one Myrtaceae outgroup, *Syzygium hemisphericum*. The release includes raw images, class labels, original image dimensions, SHA-256 file hashes, physical-specimen group identifiers, perceptual-hash component identifiers, two split manifests, and leakage-audit outputs. The two distributed split manifests comprise a specimen-stratified reference split and a strict pHash-clean group-disjoint split; both are constructed from group identifiers anchored to physical-specimen subfolders and are audited for perceptual near-duplicate leakage. The strict split contains 2025 training images, 437 validation images, and 448 test images, distributed across 56, 15, and 15 physical-specimen groups respectively, with all eight classes represented in every partition. The accompanying leakage-audit files document group overlap, exact file-hash overlap, filename overlap, and perceptual near-

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<https://doi.org/10.1016/j.dib.2026.113133>

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duplicate checks at two Hamming-distance thresholds. The dataset is intended to support reproducible benchmarking of macroscopic wood-image classifiers and to provide an example of leakage-aware data organization for biological image classification. The dataset is available at <https://doi.org/10.5281/zenodo.21216065>.

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Specifications Table

Table 1
Specifications table for the IC4SD-Wood-Eucalyptus.

Subject	Computer science; wood science; forestry
Specific subject area	Deep learning; computer vision; macroscopic wood identification; biological image classification; leakage-aware dataset evaluation
Type of data	Raw images; CSV metadata; CSV split manifests; JSON label map; leakage-audit reports; baseline-output files
How data were acquired	Prepared transverse wood surfaces were photographed at 50× magnification using a Dino-Lite Edge AM4115ZT digital microscope under controlled laboratory illumination.
Data format	Raw image files with heterogeneous original resolutions; derived CSV/JSON files for metadata, labels, split assignments, and audit outputs. No augmented images are included in the released raw-image folders.
Parameters for data collection	Images were acquired from prepared transverse cross-sections of seven <i>Eucalyptus</i> species and one <i>Syzygium hemisphericum</i> outgroup, drawn from 86 physical wood specimens (10–12 per class). Heartwood regions were prioritized when available; approximately 30 non-overlapping fields of view were targeted per specimen, except for a subset of specimens imaged under two lighting conditions (indoor artificial and outdoor natural light), which received a proportionally larger number of fields of view. Original image dimensions comprise 1280 × 1024, 2560 × 2048, and 2560 × 1920 pixels (1963, 786, and 161 images, respectively).
Description of data collection	Image files were organized into class folders with one specimen-level subfolder per physical wood specimen, checked for readability, assigned SHA-256 hashes, and mapped to physical-specimen group identifiers. Label reliability was assessed through wood-anatomical examination, specimen records, and DART-TOFMS evidence where available. For the revised release, images underwent multi-round quality control for focus, surface preparation, metadata consistency, and diagnostic-feature visibility. Affected specimens were re-imaged from the same physical wood specimens, one visually inconsistent folder was excluded, and the metadata, split manifests, and leakage-audit reports were regenerated on the final revised image set.
Data source location	IC4SD Laboratory, Posts and Telecommunications Institute of Technology, Hanoi, Vietnam Research Institute of Forest Industry (RIFI), Vietnamese Academy of Forest Sciences (VAFS), Hanoi, Vietnam
Data accessibility	Repository: Zenodo. Data identification number: 10.5281/zenodo.21216065 . Instructions for accessing these data: The Zenodo record contains raw image folders, metadata files, split manifests, leakage-audit reports, baseline-output files, README.md, LICENSE, and CITATION.cff.

1. Value of the Data

- IC4SD-Wood-Eucalyptus provides a reusable public image collection for fine-grained macroscopic identification of seven plantation-relevant *Eucalyptus* species and one Myrtaceae outgroup (*Syzygium hemisphericum*). The dataset is suitable for transfer-learning and benchmark studies in wood identification, biological image classification, and reproducible computer-vision evaluation. Because the release is distributed under CC-BY-NC-4.0, non-commercial

research applications (model development, benchmarking, and teaching) are directly supported; industrial or supply-chain timber-verification applications that constitute commercial use would require a separate licensing arrangement with the authors [1,2].

- Different user groups gain distinct value from the release. Computer-vision researchers gain a leakage-audited benchmark with predefined, group-disjoint splits for evaluating transfer-learning and reproducibility claims. Wood anatomists and forestry educators gain a curated, provenance-documented image collection covering eight anatomically similar diffuse-porous taxa, suitable for classification teaching and comparative anatomical study. Method developers working on data-leakage detection gain a documented case study of perceptual near-duplicate auditing applied to a real biological image collection.
- The release goes beyond class image folders by including per-image metadata (`metadata.csv`), SHA-256 file hashes, physical-specimen group identifiers, perceptual-hash component identifiers, and a `label_map.json` file. This structure allows users to trace every image through its class assignment, split membership, and audit record without relying on folder names or filename conventions alone [5].
- Two predefined split manifests are distributed: a specimen-stratified reference split and a strict pHash-clean group-disjoint split. Both are constructed from group identifiers anchored to the physical-specimen folder structure and are audited for perceptual near-duplicate leakage. The strict split and its associated leakage-audit files allow researchers to reproduce the conservative group-disjoint evaluation protocol and to design alternative split protocols with fully traceable audit evidence [6].
- The distributed pHash component assignments and staged leakage-audit outputs serve as a practical template for applying perceptual near-duplicate auditing to other biological image datasets. Releasing these artefacts alongside the raw images raises the reproducibility standard beyond accuracy reporting alone [5,7].
- Relative to existing public macroscopic wood-image resources such as XyloTron [1] and other CNN-based regional collections [2], IC4SD-Wood-Eucalyptus focuses on within-genus discrimination among plantation-relevant *Eucalyptus* species rather than broad cross-genus coverage, and is, to the authors' knowledge, the first macroscopic wood-image release to distribute predefined group-disjoint split manifests together with a staged, perceptual-hash-aware leakage-audit trail rather than raw images and labels alone.

2. Background

Accurate identification of timber species is a regulatory and commercial requirement in supply chains for wood products, particularly for plantation-grown genera such as *Eucalyptus* that are traded globally under shared vernacular names [1,2]. Deep-learning approaches to macroscopic wood identification increasingly rely on publicly released datasets, yet the reproducibility of reported benchmark results depends critically on how training and evaluation partitions are constructed and whether the underlying data are accessible for independent verification [5]. A common but underexamined failure mode is the presence of perceptually near-duplicate images across split boundaries—arising from repeated acquisitions, adjacent fields of view, or minor photometric variation—which can inflate held-out performance without being detected by standard filename or file-hash checks [6,7]. IC4SD-Wood-Eucalyptus was assembled specifically to address this gap: the release provides not only the image files and class labels, but also the metadata, split manifests, and staged leakage-audit reports needed to verify and reproduce the evaluation conditions.

3. Data Description

3.1. Dataset overview

The IC4SD-Wood-Eucalyptus dataset is deposited on Zenodo [10]. The dataset contains 2910 macroscopic transverse-section images drawn from 86 physical wood specimens across eight

Table 2

Physical specimens, acquisition source channels, and images per class in IC4SD-Wood-Eucalyptus. Source channels reflect the documented specimen-acquisition records available to the authors and should not be interpreted as independently verified geographic origins of the trees.

Species	Physical specimens	Specimen source/channel	Images
<i>E. camaldulensis</i>	10	Field collections, Ho Chi Minh City	288
<i>E. cladocalyx</i>	11	RIFI/VAFS collection	348
<i>E. deglupta</i>	10	Regional timber-supply source, Binh Duong	341
<i>E. diversicolor</i>	10	Regional timber-supply source, Binh Duong	337
<i>E. grandis</i>	12	Regional timber-supply source, Binh Dinh	405
<i>E. microcorys</i>	11	Regional timber-supply source, Binh Dinh	396
<i>E. saligna</i>	11	Regional timber-supply source, Binh Dinh	397
<i>S. hemisphericum</i>	11	Regional timber-supply source, Binh Dinh	398
Total	86		2910

Table 3

Class categories in the IC4SD-Wood-Eucalyptus dataset.

No.	Class name	Description
0	<i>E. camaldulensis</i>	Target species. Macroscopic transverse-section images from a plantation-relevant <i>Eucalyptus</i> species.
1	<i>E. cladocalyx</i>	Target species. Target class used for fine-grained species-level classification.
2	<i>E. deglupta</i>	Target species. Target class with visually distinctive transverse texture in the dataset.
3	<i>E. diversicolor</i>	Target species. Target class included to increase species diversity within <i>Eucalyptus</i> .
4	<i>E. grandis</i>	Target species. Widely planted plantation species with diffuse-porous transverse-section anatomy; included as a target class for fine-grained species-level classification.
5	<i>E. microcorys</i>	Target species. Australian species with durable interlocked-grain timber; transverse-section anatomy is broadly diffuse-porous and characteristic of the genus at 50× magnification.
6	<i>E. saligna</i>	Target species. Target class used for fine-grained <i>Eucalyptus</i> identification.
7	<i>S. hemisphericum</i>	Myrtaceae outgroup. Non- <i>Eucalyptus</i> outgroup from the same family, included as a single same-family outgroup reference point; not intended to characterize genus-level separability across Myrtaceae in general.

classes. Seven classes are *Eucalyptus* species: *E. camaldulensis*, *E. cladocalyx*, *E. deglupta*, *E. diversicolor*, *E. grandis*, *E. microcorys*, and *E. saligna*. The eighth class, *Syzygium hemisphericum*, is included as an outgroup from the same family, Myrtaceae. The selected *Eucalyptus* species are plantation-relevant timbers that present identification challenges under macroscopic transverse-section imaging because of broadly similar diffuse-porous anatomy within the genus [3,4]. Species names were standardized using botanical reference sources before final metadata preparation.

Each physical specimen corresponds to one subfolder in the raw image release and is identified by a stable `group_id` (Section 3.3); all images within a subfolder originate from the same physical wood block and are therefore always assigned to the same split partition. Table 1 reports the number of physical specimens and images per class. The source/channel information is provided for provenance transparency and should not be interpreted as independently certified geographic-origin or chain-of-custody evidence. A small number of specimens (identified in the metadata) were imaged under two lighting conditions – indoor artificial light and outdoor natural light – within a single physical-specimen folder; these specimens accordingly contain a larger number of images than the approximately 30 fields of view targeted per specimen elsewhere in the collection, and their lighting condition is recorded in a dedicated metadata field rather than being used to subdivide the specimen group.

Table 2 summarizes the classes and their roles in the dataset. Table 3 reports the image distribution in the strict pHash-clean split, which is the primary split distributed for conservative benchmarking. Fig. 1 provides an overview of the data preparation pipeline, and Fig. 2 shows the image distribution across the train, validation, and test partitions of the strict split.

Table 4

Image distribution across the strict pHash-clean split (Split B). Group-disjoint partitioning at the physical-specimen level yields 56 training, 15 validation, and 15 test groups. For reference, the specimen-stratified reference split (Split A) comprises 2054 training, 445 validation, and 411 test images.

Species	Train	Validation	Test	Total
<i>E. camaldulensis</i>	200	32	56	288
<i>E. cladocalyx</i>	227	96	25	348
<i>E. deglupta</i>	238	37	66	341
<i>E. diversicolor</i>	232	38	67	337
<i>E. grandis</i>	293	56	56	405
<i>E. microcorys</i>	278	59	59	396
<i>E. saligna</i>	279	59	59	397
<i>S. hemisphericum</i>	278	60	60	398
Total	2025	437	448	2910

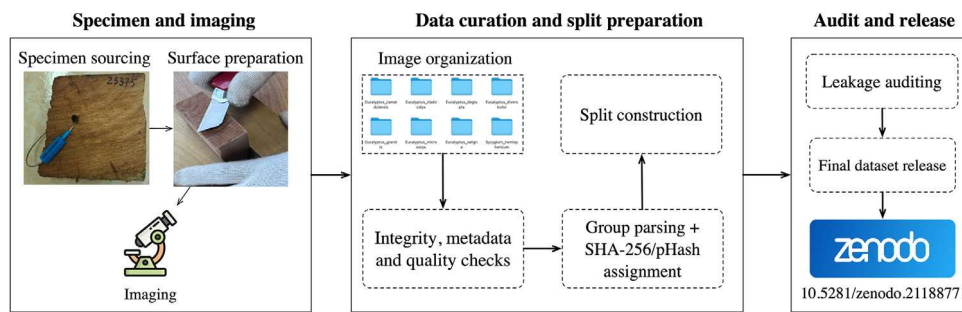


Fig. 1. Overview of the IC4SD-Wood-Eucalyptus data preparation workflow, including specimen sourcing, transverse-surface preparation, 50× imaging, image organization, integrity/metadata/quality checks, specimen-level group parsing, SHA-256 and pHash assignment, split construction, leakage auditing, and final release on Zenodo.

3.2. Repository contents

The public repository is organized to make the dataset reusable without requiring users to infer labels or split assignments from filenames alone. Table 4 lists the main files and directories.

3.3. Metadata fields

Each image is represented by one row in `metadata.csv`. The main fields are listed in Table 5. The metadata table is intended to be the authoritative index for all experiments using the dataset.

4. Experimental Design, Materials and Methods

4.1. Specimen sourcing, image acquisition, and sample preparation

IC4SD-Wood-Eucalyptus was derived from a curated wood-specimen collection assembled for macroscopic wood-identification research at IC4SD Laboratory, PTIT and RIFI, VAFS, Hanoi, Vietnam. The retained subset in this data article contains specimens representing seven *Eucalyptus* species and one *Syzygium* outgroup. The specimen acquisition was supported through institutional and collaborative collection activities under the project “Support to the implementation of the FLEGT VPA in Viet Nam”, implemented by the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH.

Table 5
Main files and folders distributed with IC4SD-Wood-Eucalyptus.

No.	File or folder and description
1	raw/ : Raw macroscopic transverse-section images organized by class folders. Each top-level folder corresponds to one species or outgroup class.
2	metadata/metadata.csv : Per-image metadata table containing image identifiers, relative paths, class labels, class indices, original dimensions, SHA-256 hashes, physical-specimen group identifiers, and pHash component identifiers.
3	metadata/label_map.json : Mapping between integer class indices and scientific class names.
4	metadata/release_counts.csv : Summary of image counts by class and release component.
5	metadata/release_manifest.csv : Release manifest used to verify file organization and to support reproducible access to the distributed image files.
6	splits/split_A_reference.csv : Specimen-stratified reference split. Physical-specimen groups, anchored to the specimen subfolders, are distributed across train/validation/test in an approximately 70/15/15 ratio.
7	splits/split_B_strict.csv : Strict pHash-clean group-disjoint split used as the primary conservative benchmark split.
8	leakage_audit/ : Group-overlap, file-hash, filename-overlap, and pHash near-duplicate audit reports for both distributed split manifests.
9	baseline_outputs/ : DenseNet-121 technical-validation outputs, including test metrics, per-class metrics, confusion matrix, training history, and generated figures.
10	README.md : Dataset description, folder structure, usage instructions, code-repository link, and citation information.
11	LICENSE : Dataset license file.
12	CITATION.cff : Citation metadata for the dataset release.
13	DATA_DICTIONARY.md : Field-level documentation for every column in every distributed CSV file, including data type, allowed values, and example entries.
14	examples/load_dataset.py : Minimal working example demonstrating how to load metadata.csv , join it with a split manifest, and instantiate a PyTorch dataset.
15	quality_control/ : Quality-control reports documenting excluded/re-imaged images, retained-image counts, sharpness scores, and representative accepted/rejected examples.

Table 6
Core metadata fields in metadata.csv.

No.	Field, type, and description
1	image_id(string) : Unique image identifier (e.g. IC4SD_EUC_000001).
2	image_path(string) : Relative path to the image (class/specimen subfolders and file name).
3	class_name(Categorical) : Species or outgroup label.
4	class_index(integer) : Numeric label from 0 to 7, consistent with label_map.json .
5	split(string) : Partition in the strict pHash-clean split (Split B): train , val , or test .
6	group_id(string) : Physical-specimen group identifier (one per specimen subfolder).
7	specimen_key(string) : Short specimen key.
8	light_condition(string) : Illumination condition: natural , indoor , or outdoor .
9	phash_component(string) : Perceptual-hash component identifier used for near-duplicate grouping.
10	constraint_component(string) : Grouping key (physical specimen unioned with perceptual component) used to construct the strict pHash-clean split.
11	sha256(string) : SHA-256 hash for exact-file duplicate detection.
12	phash(string) : 64-bit perceptual hash (hexadecimal).
13	width, height(integer) : Image dimensions in pixels.
14	source_institution(string) : Sourcing channel/institution of the specimen (e.g. RIFI/VAFS, Binh Dinh, Ho Chi Minh City); uniform within each class.
15	dart_tofms_verified(boolean) : Whether the specimen's species label was independently confirmed by DART-TOFMS (true for the eight representative specimens, one per class; false otherwise).
16	Provenance columns(string) : parsed_group_id , legacy_parsed_group_id , raw_path , original_filename , top_level_folder , and notes are retained for reproducibility and documented in DATA_DICTIONARY.md .

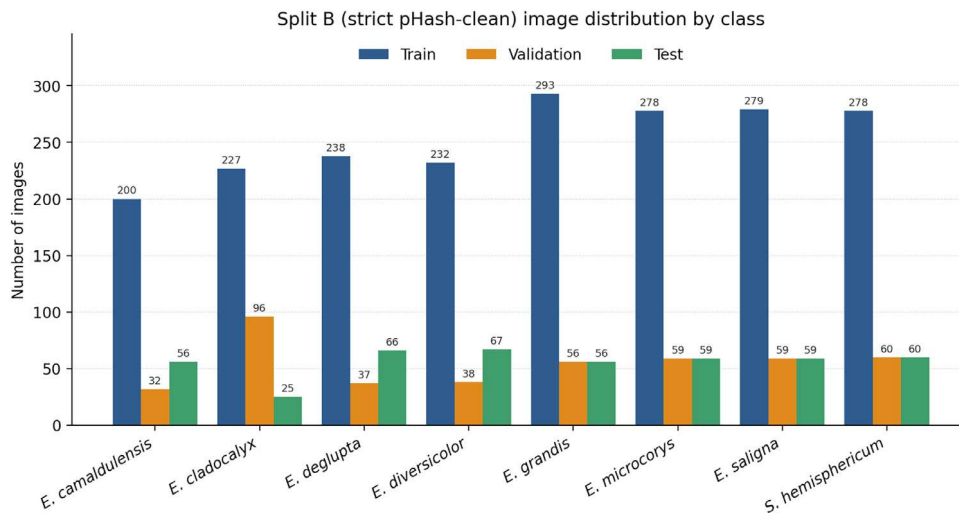


Fig. 2. Image distribution across the train, validation, and test partitions of the strict pHash-clean split (Split B).

For the revised dataset release, the retained specimens were drawn from domestic institutional and collaborative collection channels in Viet Nam, including RIFI/VAFS, regional partner timber-supply or timber-processing sources in Binh Dinh and Binh Duong, and field collections at timber-processing facilities in Ho Chi Minh City. The revised release does not include specimens obtained through international material-transfer channels. Specimen collection and original imaging were carried out in two periods: six of the eight classes were collected and imaged in October 2023, while the *E. deglupta* and *E. diversicolor* specimens were collected and imaged in November 2024. The quality-control re-imaging described in Section 4.2 (Rounds 2–3) was subsequently carried out at RIFI/VAFS in June 2026, re-imaging the same physical specimens, with the quality-control review conducted at IC4SD/PTIT. Images for any given specimen therefore originate from a single acquisition period, except for the specimens re-imaged during quality control.

All specimen transfers and use for laboratory imaging were conducted through institutional or collaborative arrangements among the participating organizations. Based on the provenance information available to the authors, the retained specimens were not documented as CITES-listed, protected, endangered, or otherwise specially regulated materials. Because the revised release consists of macroscopic images of dried wood specimens obtained through domestic institutional and collaborative channels, and does not include international transfer of genetic resources for this release, the authors are not aware of additional access-and-benefit-sharing restrictions applying to the released image dataset. Only specimens with sufficient provenance and label information were retained for imaging. This multi-source strategy was intended to introduce inter-specimen variation rather than relying on repeated images from a single physical sample.

Regarding compliance with the Convention on Biological Diversity (CBD) and the Nagoya Protocol on access and benefit-sharing: the specimens transferred between institutions are dried, morphologically prepared wood samples used exclusively for macroscopic surface imaging, and no genetic material was extracted, sequenced, or otherwise utilized.

Specimen selection criteria Specimens were selected according to documented inclusion criteria rather than by including all available material. For each target species, at least ten physical specimens were collected. Each wood sample measured at least $2 \times 3 \times 5$ cm and was taken from a log judged to be free of defects, cracks, and decay or discolouring fungi, and to be representative of the wood lot being sampled. Samples were cut from the selected logs with a saw, using clean cuts to avoid fraying or splitting, and were then labelled with a sequential sample

code and documented in a field sampling record together with their origin and other collection details. Specimens with insufficient provenance or label information were not retained.

The preparation workflow followed a standardized macroscopic wood-anatomy imaging procedure. When available, heartwood regions were selected to reduce anatomical variation caused by sapwood-heartwood transition or surface contamination. The transverse surface of each specimen was refreshed and progressively smoothed, with the final pass performed using 600-grit abrasive paper to expose vessel and parenchyma patterns. The prepared surface was slightly inclined during cutting or finishing, typically around 15–20°, a laboratory practice intended to improve the visibility of transverse cellular structures under reflected light while preserving the cross-sectional anatomical view. No formal comparative study of this inclination against a strictly perpendicular transverse cut was conducted as part of this release; the possibility of minor systematic visual effects (e.g. apparent vessel elongation) from the oblique preparation is acknowledged as a limitation (Limitations section). Only transverse cross-section images are included in this release; radial and tangential views are outside the scope of the dataset.

Images were captured using a Dino-Lite Edge AM4115ZT digital microscope at a fixed nominal magnification of 50 $\times$. Because the nominal 50 $\times$ setting denotes on-screen magnification rather than a scientific spatial scale, we report the corresponding spatial sampling directly. At this setting the horizontal field of view is approximately 9.564 mm (± 0.192 mm) and is held constant across capture modes, so the pixel pitch depends only on image width: about 7.47 $\mu\text{m}/\text{pixel}$ for the 1280 $\times$ 1024 images and about 3.74 $\mu\text{m}/\text{pixel}$ for the 2560 $\times$ 2048 and 2560 $\times$ 1920 images. The per-image pixel dimensions recorded in `metadata.csv` allow the exact spatial scale of any image to be recovered.

During acquisition, the prepared wood surface was kept under controlled indoor illumination for the majority of specimens; a subset of specimens (identified via the `light_condition` metadata field) were additionally imaged outdoors under natural light within the same physical-specimen folder. Indoor illumination was provided by the digital microscope's integrated LED light source, while the outdoor subset was imaged under natural daylight; within each lighting condition, illumination was kept consistent across specimens and imaging sessions. All specimens were dried wood samples (moisture content below 20%). The colour temperature of the integrated LED and of the outdoor daylight was not separately recorded or calibrated, so any residual illumination variation is documented only through the `light_condition` metadata field. The microscope was positioned to capture non-overlapping fields of view across each specimen whenever possible. Approximately 30 fields of view were targeted per specimen, depending on specimen size and surface quality, with specimens imaged under two lighting conditions receiving proportionally more. The raw files retain the microscope-specific acquisition record, including heterogeneous image dimensions. The released set spans three native resolutions—1280 $\times$ 1024 (1963 images), 2560 $\times$ 2048 (786 images), and 2560 $\times$ 1920 (161 images)—reflecting the use of more than one acquisition and export configuration across collection periods. These original resolutions are documented in `metadata.csv` rather than being hidden by forced resizing in the released raw-image folders.

Species labels were checked before final inclusion using wood-anatomical examination of macroscopic transverse-section features and available reference information. As an additional label-reliability check, DART-TOFMS [11] chemical-fingerprint evidence was available for one representative physical specimen per class, i.e., 8 of the 86 specimens in the released dataset. These records were used as supplementary evidence alongside wood-anatomical examination and specimen records, but they should not be interpreted as complete chemical verification of every specimen in the dataset. The raw DART-TOFMS [11] records are not distributed with this macroscopic-image dataset because they are held by collaborating institutions and fall outside the scope of the released image-data package. Specimens with inconsistent or insufficient identification evidence were excluded from the final dataset release. The resulting data package therefore combines image files with label, source, grouping, and audit metadata so that users can trace each image through the released manifests and validation files.

The released image files preserve their original acquisition dimensions. During model training, users may resize images to the input resolution required by a given architecture; how-

ever, such resizing is not applied to the raw images in the repository. This design preserves the acquisition record while allowing different research groups to apply their own preprocessing pipelines.

4.2. Image integrity and multi-round quality control

Before metadata construction, all image files were checked for readability using Pillow. Files were retained only if they could be opened and decoded successfully. Original width and height were recorded for every image. SHA-256 file hashes were computed to detect exact duplicates independently of filename. The released dataset does not include augmented images; augmentation, if used, should be applied only at training time. This readability check is a minimal integrity filter and does not by itself assess visual quality (focus, surface preparation, or the presence of diagnostic anatomical features); the dedicated quality-control process described below addresses this directly.

For the revised dataset release, we conducted a dedicated dataset-quality assessment focusing on image focus, specimen-surface preparation, metadata consistency, and the visibility of macroscopic wood-anatomical features. The assessment proceeded in three rounds:

- **Round 1 – identification.** A systematic visual review of the candidate release identified that a subset of images from five physical specimens (*E. cladocalyx* specimen - 3282, *E. grandis* specimen - 3284, *E. saligna* specimen - 3293, *E. camaldulensis* specimen - 3358, and *S. hemisphericum* specimen 3297) had been acquired from transverse surface regions that were insufficiently prepared or did not consistently display visible vessel and parenchyma structure. In addition, one *E. grandis* image folder containing 40 images was visually inconsistent with the remainder of the class and was excluded from the revised release.
- **Round 2 – re-imaging and re-verification.** Because the affected physical specimens were retained in the authors' collection, they were re-imaged from newly prepared transverse surface regions of the same physical specimens under the acquisition protocol described in [Section 4.1](#). The affected images were therefore replaced by newly acquired images from the same specimens rather than by images from different specimens. Manual quality-control review of this round flagged 296 candidate images for exclusion and re-imaging because of image-quality issues or filename/metadata inconsistencies.
- **Round 3 – final re-imaging and consistency verification.** The 296 flagged candidate images were re-imaged, and the complete revised image set was checked against the metadata table using an automated consistency script. The script compared (i) the number of species folders and physical-specimen folders, (ii) per-specimen and per-class image counts, (iii) file-type uniformity, and (iv) duplicate filenames across folders. The final revised release passed this consistency check, with the on-disk folder structure and metadata table agreeing on 2910 released images from 86 physical specimens.

Representative examples from the quality-control review are shown in [Fig. 3](#). The retained example shows a prepared transverse surface with visible vessel and parenchyma structure, whereas the excluded example illustrates an insufficiently prepared or non-diagnostic field of view that was rejected during the revised quality-control process.

To complement the manual quality-control review, we also quantified image sharpness across the full revised release. For each of the 2910 released images, we computed the variance of the image Laplacian as a no-reference focus measure, where larger values indicate stronger local high-frequency structure. Before computing the score, images were converted to grayscale and resized to a common longer side of 1024 pixels to reduce the effect of heterogeneous native resolutions on the sharpness statistic.

Although no single Laplacian-variance threshold is universal across imaging systems, resolutions, and wood textures, the resulting distribution provides a reproducible screen for low-sharpness images. The median score was 555.3, with a 5th percentile of 158.6 and a 1st percentile of 98.8. Only 2 images (0.07%) scored below 50, and 32 images (1.10%) scored below 100.

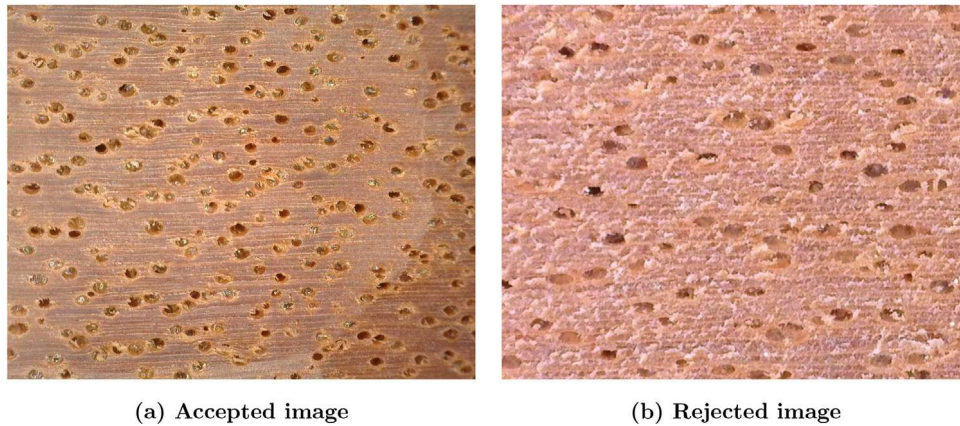


Fig. 3. Representative accepted and rejected images from the revised quality-control process for *E. cladocalyx* specimen 3282. The accepted image was retained because the prepared transverse surface displays visible macroscopic wood-anatomical structure. The rejected image was excluded because the field of view did not consistently show diagnostic vessel and parenchyma features after quality-control review.

Class-level medians ranged from 271.2 for *E. cladocalyx* to 1,172.0 for the *Syzygium hemisphericum* outgroup. These results indicate that the retained release contains only a small low-sharpness tail after the multi-round manual quality-control process. The per-image sharpness scores are distributed with the dataset so that users may apply stricter thresholds if required for a specific downstream task.

Beyond focus, the manual quality-control review also assessed whether each field of view displayed the macroscopic diagnostic features relevant to identification—principally the vessel arrangement and axial-parenchyma patterns visible in transverse section at 50× magnification. Fields of view that did not clearly show these structures were flagged for exclusion and re-imaging. Of the 296 images flagged in Round 2, 241 (all from *E. cladocalyx* specimen 3282) were rejected specifically because the imaged surface region did not display diagnostic vessel and parenchyma structure, and were subsequently re-imaged from newly prepared surface regions of the same specimen; the remaining flagged images were removed for focus or filename/metadata issues. The retained release therefore consists of transverse-section fields of view in which these macroscopic diagnostic features are visible, as illustrated by the accepted example in Fig. 3.

Across the three quality-control rounds, one *E. grandis* image folder (40 images) was excluded outright, and a further 296 images were flagged for image-quality or filename/metadata issues and replaced by newly acquired images from the same physical specimens; these flagged images correspond to approximately 10% of the released set. The final retained release comprises 2910 images from 86 physical specimens. The exclusion and replacement history, including the reasons for exclusion and the final retained-image counts, is documented in the released quality-control report files. Duplicate and cross-split overlap checks are documented separately in the leakage-audit files.

4.3. Dataset splitting

Two split manifests are included to support different evaluation strictness levels. Both splits are constructed as group-disjoint partitions over the physical-specimen `group_id` (Section 3.3): every image belonging to a given physical specimen is assigned to exactly one of the train, validation, or test partitions. This corrects the original release, in which group identifiers were parsed from filenames and, for several classes, incorrectly merged multiple distinct physical specimens into a single group (Section 4.2). In the revised release, `group_id` is defined directly as the name of the physical-specimen subfolder (one folder per specimen), so no filename

Table 7

Per-class pHash component counts and component-size statistics for IC4SD-Wood-Eucalyptus. Components were computed within each class using a pHash Hamming-distance threshold of 10.

Species	Images	Components	Median size	Mean size	Max size
<i>E. camaldulensis</i>	288	288	1	1.00	1
<i>E. cladocalyx</i>	348	339	1	1.03	3
<i>E. deglupta</i>	341	340	1	1.00	2
<i>E. diversicolor</i>	337	336	1	1.00	2
<i>E. grandis</i>	405	402	1	1.01	2
<i>E. microcorys</i>	396	377	1	1.05	6
<i>E. saligna</i>	397	388	1	1.02	3
<i>S. hemisphericum</i>	398	374	1	1.06	3
Overall	2910	2844	1	1.02	6

parsing or regular-expression extraction is involved; the earlier filename-derived identifiers are retained only in the `legacy_parsed_group_id` field for provenance.

Split A is the specimen-stratified reference split. Physical-specimen group identifiers are distributed across partitions in approximately a 70/15/15 ratio using `GroupShuffleSplit` from `scikit-learn` [8]. Split A comprises 2054 training, 445 validation, and 411 test images (2910 in total).

Split B is the strict pHash-clean group-disjoint split. Within each class, perceptual hashing [7] was used to identify visually near-duplicate images. Images connected by pHash Hamming distance ≤ 10 were merged into the same perceptual component; because perceptual near-duplicates typically originate from the same physical specimen, these components are consistent with, and further refine, the physical-specimen grouping used for Split A. Components, rather than individual images, were then assigned to train, validation, and test partitions using `GroupShuffleSplit` [8]. Split B comprises 2025 training, 437 validation, and 448 test images, distributed across 56 training, 15 validation, and 15 test groups; the per-class distribution is reported in Table 3.

Using the pHash Hamming-distance threshold of 10, the 2910 images formed 2844 within-class pHash components (Table 6). The component-size distribution was highly sparse, with a median of 1 image, a mean of 1.02 images, and a maximum of 6 images per component. This indicates that most images were isolated under the near-duplicate criterion used to construct the strict split, while the few multi-image components were kept within the same partition during split construction.

Choosing a split. We recommend Split B (the strict pHash-clean split) as the default for reporting benchmark and reproducibility results, because it removes perceptual near-duplicates prior to partitioning and therefore gives the most conservative estimate of generalization to unseen specimens. Split A (the specimen-stratified reference split) is provided for two purposes: as a compatibility baseline for comparison with pipelines that enforce only specimen-level disjointness, and as a sensitivity reference for quantifying how much a less strict, non-pHash-cleaned partition affects reported performance. Because Split A retains a small number of perceptual near-duplicate pairs across partitions (two at Hamming distance ≤ 10 ; Table 7), results obtained on Split A may be modestly optimistic relative to Split B and should not be reported as strictly leakage-controlled. Studies that compare against results reported here should state explicitly which split was used.

4.4. Leakage-audit procedure

A staged leakage audit was applied to both Split A and Split B, following practices for dataset-level leakage detection [6]. Because exact file-hash (SHA-256) comparison detects only byte-identical files and cannot, on its own, identify visually overlapping or near-duplicate fields of view, the audit combines exact-hash checks with perceptual hashing, which is sensitive to such visual near-duplication; the fields of view were, moreover, acquired to be non-overlapping by

Table 8

[Revised] Leakage-audit summary for the reference split (Split A) and the strict pHash-clean split (Split B). Each entry reports the number of cross-partition violations detected by the corresponding check (0 indicates none detected). The identical five-stage audit was applied to both splits. Split A retains two perceptual near-duplicate pairs at Hamming distance ≤ 10 , consistent with its role as a less conservative reference split, whereas Split B shows no detected leakage under any check.

Audit item	Split A	Split B
Group overlap	0	0
Exact cross-split SHA-256 hash overlap	0	0
Filename overlap	0	0
pHash cross-split pairs at Hamming ≤ 5	0	0
pHash cross-split pairs at Hamming ≤ 10	2	0

design during imaging (Section 4.1). In the original release, only Split B was audited for perceptual near-duplicates; this revision applies the identical five-stage audit to Split A as well, since correcting the specimen-grouping error (Section 4.2) also required re-evaluating whether the reference split contains cross-partition near-duplicates. The released audit files include group overlap, exact SHA-256 file-hash overlap, filename overlap, and pHash near-duplicate checks at Hamming-distance thresholds of 5 and 10, for both splits. In addition to the pass/fail summary, the `leakage_audit/` folder now distributes the full list of any cross-partition candidate pairs identified at each threshold (not only zero-finding summary counts), so that the audit trail is independently inspectable rather than only asserted. Feature nearest-neighbour inspection was used as an additional manual-review aid during dataset checking: an ImageNet-pretrained ResNet-50 (final classification layer removed) was used to extract image embeddings, and for each image the five nearest neighbours by cosine similarity were compiled into contact sheets for manual visual review; this step does not apply a fixed similarity threshold and is not treated as an automatic pass/fail leakage test. Table 7 summarizes the audit outcome for both distributed splits.

As summarized in Table 7, the strict pHash-clean split (Split B) contains no detectable cross-partition leakage under any of the applied checks, while the specimen-stratified reference split (Split A) is likewise free of exact-duplicate and filename collisions and contains only two residual perceptual near-duplicate pairs at Hamming distance ≤ 10 (none at ≤ 5). This small residual is expected: Split A is provided as a less conservative, specimen-stratified baseline, whereas Split B additionally merges perceptual near-duplicate components prior to partitioning and therefore exhibits no detectable near-duplicate leakage at any threshold. Both audit trails are distributed so that users can select the split matching their intended evaluation strictness.

4.5. Baseline performance for wood species classification

A DenseNet-121 [9] classifier was trained as a lightweight technical-validation baseline to verify that the released image files, labels, class mapping, and strict split manifest can be used in an end-to-end transfer-learning workflow. This experiment was not designed as a comprehensive model benchmark.

The baseline used the strict pHash-clean split (Split B), comprising 2025 training images, 437 validation images, and 448 test images. The model was initialized with ImageNet-pretrained weights. Images were resized to 224×224 pixels and normalized using ImageNet statistics. The final DenseNet-121 classification layer was replaced with an eight-class output head. Training used AdamW with a learning rate of 1×10^{-4} , weight decay of 1×10^{-2} , cross-entropy loss with label smoothing of 0.1, a batch size of 64, automatic mixed precision on CUDA, and seed 42. The run was trained for 50 epochs, and the best checkpoint was selected according to validation macro-F1. The run was conducted on Linux 6.6.122+ using Python 3.12.13, PyTorch 2.11.0+cu128, torchvision 0.26.0+cu128, scikit-learn 1.6.1, CUDA 12.8 runtime, and an NVIDIA A100-SXM4-40GB GPU.

Table 9

DenseNet-121 technical-validation results on the strict pHash-clean split (Split B). The baseline was used to verify dataset usability rather than to provide a comprehensive model benchmark.

Metric	Value	Notes
Accuracy (%)	95.76	Strict Split B test set (429/448)
Macro precision (%)	94.20	Eight-class macro average
Macro recall (%)	95.82	Eight-class macro average
Macro-F1 (%)	94.52	Main summary metric for the technical-validation run
Weighted F1 (%)	95.82	Weighted by class support
Test loss	0.5587	Cross-entropy loss with label smoothing of 0.1
Best epoch	18	Checkpoint selected by validation macro-F1
Best validation macro-F1 (%)	99.78	Validation score at the selected checkpoint
Training setting	ImageNet-pretrained DenseNet-121	50 epochs, batch size 64, AdamW, learning rate 1×10^{-4} , weight decay 1×10^{-2} , seed 42

Table 10

Per-class precision, recall, F1-score, and test-set support for the DenseNet-121 technical-validation baseline on the strict pHash-clean split (Split B).

Species	Precision	Recall	F1	Support
<i>E. camaldulensis</i>	1.0000	1.0000	1.0000	56
<i>E. cladocalyx</i>	0.6757	1.0000	0.8065	25
<i>E. deglupta</i>	1.0000	1.0000	1.0000	66
<i>E. diversicolor</i>	1.0000	1.0000	1.0000	67
<i>E. grandis</i>	0.8936	0.7500	0.8155	56
<i>E. microcorys</i>	0.9667	0.9831	0.9748	59
<i>E. saligna</i>	1.0000	0.9322	0.9649	59
<i>S. hemisphericum</i>	1.0000	1.0000	1.0000	60
Macro average	0.9420	0.9582	0.9452	448

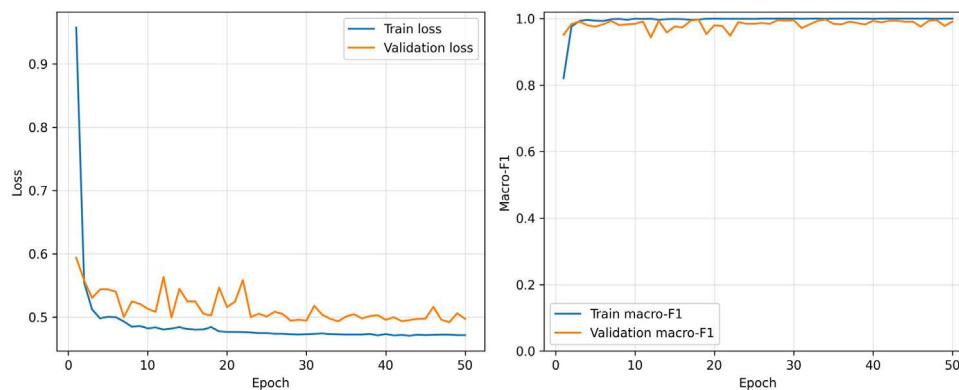


Fig. 4. Training and validation curves for the DenseNet-121 technical-validation baseline on the strict pHash-clean split (Split B).

On the held-out strict test set, DenseNet-121 achieved 95.76% accuracy, 94.20% macro precision, 95.82% macro recall, 94.52% macro-F1, and 95.82% weighted F1 (Table 8).

The baseline results are summarized in Table 8, with per-class metrics in Table 9, training curves in Fig. 4, and the confusion matrix in Fig. 5. Four of the eight classes were classified without error (*E. camaldulensis*, *E. deglupta*, *E. diversicolor*, and *S. hemisphericum*, all at F1 = 1.0000).

The 19 test-set errors centre on *E. grandis*: 12 of its 56 test images were predicted as *E. cladocalyx* and a further 2 as *E. microcorys*, lowering *E. grandis* recall to 0.7500, while 4 *E. saligna* and 1 *E. microcorys* image were in turn predicted as *E. grandis*. Consequently *E. cladocalyx* pre-

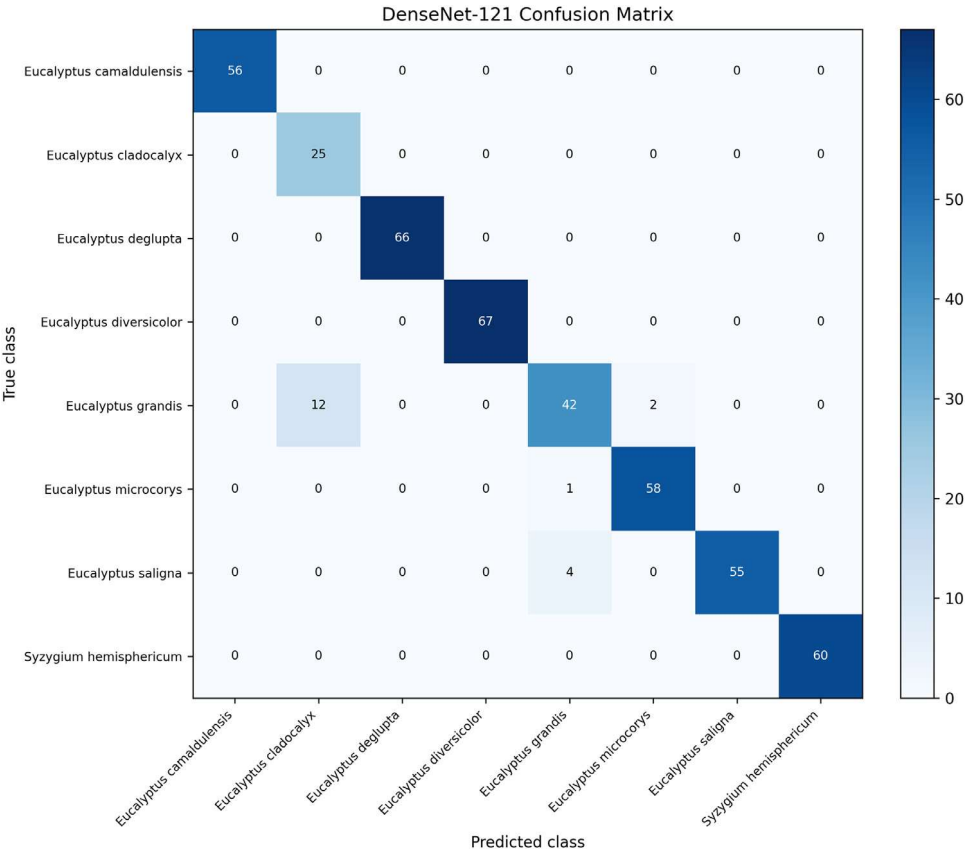


Fig. 5. Confusion matrix of the DenseNet-121 technical-validation baseline on the strict pHash-clean (Split B) test set.

cision falls to 0.6757 (it absorbs 12 false positives while retaining perfect recall) and *E. grandis* precision to 0.8936. These confusions are consistent with the broadly similar diffuse-porous transverse-section anatomy shared among these plantation *Eucalyptus* species at 50× magnification, where vessel arrangement and axial-parenchyma patterns overlap substantially [3,4]. The misclassifications are moreover concentrated in a small number of specimen groups rather than spread evenly across each class, indicating a few challenging specimen-level cases rather than broad label confusion.

As this run is a single-seed technical-validation baseline rather than a comprehensive benchmark, and several classes have small per-class test supports (25–67 images), the per-class metrics should be read as indicative rather than definitive. The best validation macro-F1 (99.78%, computed over only 15 held-out groups with one to two specimens per class) exceeds the test macro-F1 by roughly 5.3 percentage points; this gap reinforces that the baseline is intended to establish dataset usability under a standard transfer-learning workflow rather than to bound achievable performance.

Limitations

Although IC4SD-Wood-Eucalyptus provides a controlled and leakage-audited macroscopic image resource, several limitations should be considered:

- **Species coverage limited to the current laboratory collection.** The dataset contains images from seven *Eucalyptus* species and one *Syzygium hemisphericum* outgroup selected from specimens available at IC4SD Laboratory and partner institutions. This subset does not represent the full biological or commercial diversity of the genus or of Myrtaceae, and conclusions drawn from models trained on this collection are specific to its eight classes.
- **Moderate class imbalance and uneven validation/test support.** Per-class image counts range from 288 (*E. camaldulensis*) to 405 (*E. grandis*). Because both split manifests are group-disjoint at the physical-specimen level, per-class validation and test sizes are determined by whole-specimen assignment rather than balanced by image count. In the strict split (Split B) this produces an uneven distribution for some classes—most notably *E. cladocalyx*, whose dual-lighting specimen falls into the validation partition (96 validation versus 25 test images)—and per-class test supports range from 25 to 67 images. Per-class metrics for classes with small test support should therefore be read as indicative, and users comparing models may prefer macro-averaged metrics (as reported for the baseline) or specimen-aware resampling.
- **Multi-site acquisition with heterogeneous lighting.** Images were captured primarily at IC4SD Laboratory (PTIT) and RIFI using the same microscope model under standardized indoor conditions; a subset of specimens were additionally imaged outdoors under natural light. Users who apply the dataset to train models intended for other imaging environments should be aware that cross-device, cross-laboratory, and broader field-condition generalization have not been evaluated as part of this data release, and that the indoor/outdoor lighting subset (documented in the `light_condition` field) introduces a modest, documented acquisition-condition heterogeneity that has not been formally assessed for batch effects.
- **Heterogeneous original resolutions.** The raw images preserve heterogeneous acquisition resolutions, including some lower-resolution files. Users should report any resizing or filtering decisions when reusing the dataset.
- **Group identifiers depend on curated folder organization.** The distributed `group_id` values are anchored to physical-specimen subfolders established during data curation, correcting a specimen-merging error present in the filename-parsed identifiers of the original release. While this is a substantial improvement in specimen-level traceability, the folder-level organization itself was assigned manually during curation and was not independently re-verified by a third party; users requiring formally audited chain-of-custody records should treat `group_id` as curator-asserted rather than externally certified.
- **Leakage auditing is conservative but not exhaustive.** Perceptual near-duplicate removal under fixed Hamming thresholds eliminates identifiable cross-partition near-duplicates; however, images from the same physical specimen taken under slightly different conditions may remain perceptually distinct while sharing acquisition-specific texture characteristics that strict pHash thresholding cannot detect.
- **No microscopic or pixel-level annotations.** The dataset provides macroscopic transverse-section images and species-level class labels. Cell-level anatomical segmentation, microscopic measurements, and pixel-level anatomical annotations are outside the scope of this release.

Ethics Statement

This study did not involve human participants, animal subjects, social-media data, or personal data. The dataset consists only of laboratory images of dried, morphologically prepared wood specimens; no living material was collected, and no genetic material was extracted, sequenced, or otherwise used.

The specimens were provided through RIFI/VAFS, Viet Nam, and institutional collection activities associated with the GIZ-implemented FLEGT VPA project (Section 4.1). Based on the provenance information available to the authors, the retained specimens were not documented as CITES-listed, protected, endangered, or otherwise specially regulated materials. Their use for laboratory imaging followed the relevant institutional procedures of the participating organizations.

The authors are not aware of unresolved access-and-benefit-sharing restrictions affecting the release of the macroscopic image dataset.

Recommended Reporting and Versioning Policy

For comparable benchmarking, users should report the split used (Split A or Split B), any additional filtering or preprocessing, random seed(s), and evaluation metric(s). This checklist is also provided in the repository README.md.

Future corrections to labels, metadata, split manifests, or audit files will be released as new versioned Zenodo records, not by silently modifying archived versions. Version differences are documented in CHANGELOG.md; the Zenodo concept DOI points to the latest version, while each version-specific DOI remains citable.

Code Availability

Scripts for metadata checking, split construction, leakage auditing, and the DenseNet-121 technical-validation experiment are available at <https://github.com/khanhnt/IC4SD-Wood-Eucalyptus>. The repository is publicly accessible, released under the MIT License for code, and includes a tagged release corresponding to the archived Zenodo dataset version.

CRediT Author Statement

Khanh Nguyen-Trong: Conceptualization; Methodology; Software; Validation; Formal analysis; Data curation; Investigation; Visualization; Writing – original draft; Writing – review & editing; Project administration. **Trinh Nguyen-Thi:** Data curation; Investigation; Resources; Writing – review & editing. **Nghia Nguyen-Trong:** Data curation; Investigation; Resources; Writing – review & editing.

Funding

The preparation and revision of this data article did not receive specific external funding. The underlying specimen acquisition and data-collection activities were partly supported through institutional and collaborative activities under the project “Support to the implementation of the FLEGT VPA in Viet Nam”, implemented by the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH.

Declaration on the use of Generative AI

Generative AI assistance was used for language editing, manuscript-structure organization, and polishing suggestions. All dataset design decisions, specimen preparation, imaging, metadata construction, split generation, leakage-audit outputs, and experimental results remain the responsibility of the authors. No generative AI tool was used to create, modify, or fabricate any image file, metadata value, audit result, or numerical result reported in this article.

Data Availability

IC4SD-Wood-Eucalyptus: A macroscopic transverse-section image dataset with metadata, split manifests, and leakage-audit reports for *Eucalyptus* wood identification (Original data) is available at **10.5281/zenodo.21216065**. The dataset is released under **CC-BY-NC-4.0**.

Acknowledgements

The authors thank the Intelligent Computing for Sustainable Development Laboratory (IC4SD), Posts and Telecommunications Institute of Technology (PTIT), and the Research Institute of Forest Industry (RIFI), Vietnamese Academy of Forest Sciences (VAFS), for supporting specimen preparation, imaging, label checking, and computational experimentation.

Declaration of Competing Interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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